BRIDGES IN MATHEMATICS • GRADE 5 • UNIT 2

Adding & Subtracting Fractions with Unlike Denominators

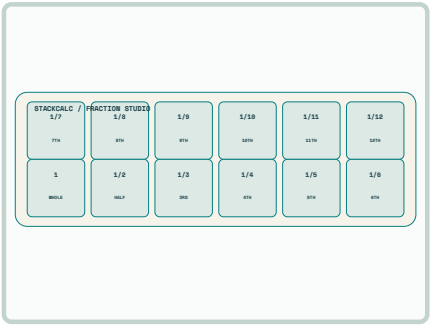
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.5.NF.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Fraction Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place 1/2 block (42mm) and 1/3 block (28mm) end to end. Height is 70mm.

Sketch Model on Paper:
Draw an 84mm bar partitioned into sixths: shade 3 sixths (1/2) and 2 sixths (1/3).

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	FDISP	Fraction Studio	
1.	1/2	X: 1/2	Key 1/2 (.1.2) into
2.	ENTER	Y: 1/2 | X: 1/2	Stack lift
3.	1/3	Y: 1/2 | X: 1/3	Key 1/3 (.1.3) into
4.	+	X: 5/6	Add & common denom:

EXPECTED LCD X: 5/6 (0.8333)

1 **Add 1/4 + 1/3 using Fraction Studio.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Add 2/3 + 1/6 using Fraction Studio.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Subtract 3/4 - 1/2 using Fraction Studio.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Subtract 5/6 - 1/3 using Fraction Studio.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 2

Adding & Subtracting Fractions with Unlike Denominators • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Fraction Studio Towers (84.0mm Whole)

CAD Blueprint: fraction_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/fraction_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal 3x1 Bin & Color-Coded Towers (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Add 1/4 + 1/3 using Fraction Studio.** Standard: CCSS.MATH.5.NF.A.1	7/12	[. 1 . 4] [ENTER] [. 1 . 3] [+]
2	**Add 2/3 + 1/6 using Fraction Studio.** Standard: CCSS.MATH.5.NF.A.1	5/6	[. 2 . 3] [ENTER] [. 1 . 6] [+]
3	**Subtract 3/4 - 1/2 using Fraction Studio.** Standard: CCSS.MATH.5.NF.A.1	1/4	[. 3 . 4] [ENTER] [. 1 . 2] [-]
4	**Subtract 5/6 - 1/3 using Fraction Studio.** Standard: CCSS.MATH.5.NF.A.1	1/2 (3/6)	[. 5 . 6] [ENTER] [. 1 . 3] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Students must not add denominators directly (e.g. 1/2 + 1/3 is NOT 2/5).

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.